

Nicolás Aller Ponte

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Education

UDC · USC · CESGA, MSc in High Performance Computing (HPC) – A Coruña, Spain Sept 2026 - present

- 60-ECTS joint master's degree: high performance architectures, parallel programming (OpenMP, MPI), heterogeneous architectures (CUDA) and HPC in the cloud, with lab work on CESGA's Finisterrae supercomputer.

Universidade da Coruña, Bachelor's Degree in Data Science and Engineering – A Coruña, Spain Sept 2022 - July 2026

- Knowledge of the complete data lifecycle: HPC, statistics, data engineering, databases and AI.
- Bachelor's Thesis graded 10/10 with Honors (Matricula de Honor): Computational Optimization of ReliefF, strategies based on approximate search and prototyping.

Experience

Data & AI Engineer, Pull&Bear (Inditex) – A Coruña, Spain Sept 2026 - present

- Design and implementation of data and AI solutions for the distribution department.
- End-to-end LLM-powered agents and automations for department tasks.
- Data modeling and serving in Snowflake and Power BI dashboards to track key indicators, versioned and deployed through Azure DevOps.

Freelance Developer, Self-employed – A Coruña, Spain May 2026 - July 2026

- End-to-end design and implementation of SGAR, an in-house product automating Sergas prescription management in pharmacies.

Research Assistant, Inditex-UDC Chair of AI in Green Algorithms – A Coruña, Spain Nov 2025 - Mar 2026

- Research and development of efficient algorithms for AI models.

AI Engineer (Internship contract), Gradient – A Coruña, Spain Sept 2025 - Oct 2025

- Development of a car damage detection and parts evaluation system.

Languages

Proficiency: Spanish (Native), Galician (Native), English (C1, EF SET)

Skills

Data Engineering & Cloud: Python, R, SQL, Snowflake, Power BI, MongoDB, Docker, Git, Spark, Azure DevOps, AWS, GCP

AI & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, NLP, LLMs & agents

HPC & Systems Programming: C, C++, Parallel Computing (OpenMP, MPI), CUDA

Awards

Inditex Tech Ambassador: Selected for the Inditex Tech Ambassador Program, representing Inditex Tech at universities and tech events.

First Place - Best Use of Open Source AI (HackUDC 2026): Recognition for the integration and optimization of open-source models (SigLIP, DINOv2) in a scalable production environment.

Second Place - Inditex-Tech Challenge (HackUDC 2026): Finalist for developing a visual search engine for massive fashion catalogs.

Second Place - GDG Challenge (Impacthon 2026): Runner-up for developing LocalFold, a web portal for protein structure prediction using AlphaFold2, A2A multi-agent system, and Gemini-powered analysis on Google Cloud.

Projects

SGAR - Sergas Prescription Automation – A Coruña, Spain May 2026

- End-to-end system that automates Sergas prescription handling in pharmacies, replacing a multi-hour daily manual workflow.
- GDPR/LOPD compliance for medical data (anonymous logging, strict exclusion of personal health information).

Outfit Intelligence - HackUDC – A Coruña, Spain Mar 2026

- Multi-layer visual search pipeline using YOLOv8 and SigLIP-384 to identify catalog products in outfit images, with model fine-tuning.
- Recommendation engine using FAISS and a REST API with FastAPI on a catalog of 27,000+ products.

LocalFold - Impacthon 2026 – GDG Challenge Apr 2026

- Web portal for protein structure prediction with AlphaFold2 without HPC knowledge, with interactive 3D visualization (Mol*).
- A2A multi-agent system with four specialized agents using Gemini with Google Search Grounding.
- Deployed on Google Cloud Run with FastAPI, Firestore, and multilingual support.

Hobbies

Football Coach 2022 - present

- Coaching youth teams of children between 6 and 11 years old.